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Talha Saleem

Education

Sep-2022 – Jul-2026 **Quaid-i-Azam University**, Islamabad, Pakistan.
- B.S. in Bioinformatics
- CGPA: 3.0/4.0

Oct-2021 – Oct-2021 **Govt. Rizvia Islamia Degree College**, Haroonabad, Pakistan.
- Pre-Medical (F.Sc.)
- Marks: 92%

Apr-2019 – Apr-2019 **Govt. Model High School**, Haroonabad, Pakistan.
- Secondary School Certificate (Matric) Biology, Mathematics
- Marks: 94.7%

Selected Projects

2026 **BriefAI: Real-Time Meeting Transcription and Multilingual Summarization Platform**.
Talha Saleem
Tools: FastAPI, WebSockets, faster-whisper, Ollama (Qwen3-1.7B, Llama 3.2-1B), SpeechBrain, React, JWT
- Developed a full-stack web app with a React frontend and FastAPI backend hosting AI services.
- Integrated AI services for fine-tuning, inference, text generation, and data engineering.
- Built a live streaming pipeline using FastAPI WebSockets and a browser AudioWorklet capture.
- Transcribed microphone audio in real time using faster-whisper for speech-to-text.
- Implemented speaker diarization using SpeechBrain ECAPA-TDNN embeddings.
- Clustered speaker embeddings using agglomerative clustering for diarization.
- Integrated Qwen3-1.7B and Llama 3.2-1B via Ollama for summarization and translation.
- Extracted action items and generated Markdown notes from meeting transcripts.
- Secured the platform using JWT-based authentication and API rate limiting.
- Benchmarked Qwen3-1.7B against Llama 3.2-1B on latency and token throughput.

2026 **ResumeRanker: AI Resume Parser and Candidate Matching Platform**.
Talha Saleem
Tools: Python, spaCy, TF-IDF/BM25, Docker, PostgreSQL
- Built an NER pipeline using spaCy to parse resumes in PDF and DOCX formats.
- Extracted skills, education, experience, projects, and certifications automatically.
- Designed a TF-IDF/BM25 algorithm for candidate-job relevance matching.
- Applied skill normalization and weighted scoring across job descriptions.
- Auto-tagged candidates across backend, frontend, and full-stack profiles.
- Auto-tagged candidates across data science, AI/ML, and bioinformatics profiles.
- Deployed a Dockerized FastAPI backend for parsed resumes and job posts.
- Stored ranking scores and explainable match logs using PostgreSQL.

2026 **BioSearchAI: Biomedical RAG and Semantic Literature Search Platform**.
Talha Saleem
Tools: BioBERT, Sentence Transformers, FAISS, PubMed API
- Built a document pipeline for PubMed abstracts and research PDFs.
- Performed text extraction, chunking, metadata parsing, and cleaning.
- Fine-tuned BioBERT for biomedical entity extraction across genes and diseases.
- Extended entity extraction to drugs, proteins, organisms, and clinical terms.
- Integrated Sentence Transformers with FAISS for semantic vector retrieval.
- Retrieved sentence-level evidence from biomedical abstracts and documents.
- Evaluated extraction quality using precision, recall, and F1-score.
- Evaluated retrieval quality using top-k accuracy and manual error analysis.

Work Experience

Software Engineer, Freelance, Islamabad, Pakistan.

Tools: FastAPI, SQLAlchemy, Docker, React, TypeScript, WebSockets

- Designed and built REST and WebSocket APIs using FastAPI for real-time data streaming applications.
- Modeled relational databases with SQLAlchemy ORM and managed versioned schema migrations.
- Implemented JWT-based authentication with refresh tokens and rate-limited API endpoints.
- Containerized backend services using Docker with Nginx as a reverse proxy for deployment.
- Built responsive frontends using React and TypeScript, integrated with backend APIs via axios.
- Developed a browser AudioWorklet module for real-time audio capture and streaming.
- Validated application functionality using Vitest and pytest across frontend and backend layers.

AI Engineer, South Korea, Remote.

Tools: Ollama, faster-whisper, spaCy, BioBERT, Sentence Transformers, FAISS

- Deployed open-weight LLMs (Qwen3-1.7B, Llama 3.2-1B) via Ollama for summarization and translation.
- Built speech-to-text pipelines using faster-whisper and speaker diarization using SpeechBrain.
- Designed NER and ranking pipelines using spaCy and TF-IDF/BM25 for text classification tasks.
- Fine-tuned BioBERT using Hugging Face Transformers and PyTorch for biomedical entity extraction.
- Built semantic retrieval systems using Sentence Transformers and FAISS over biomedical literature.
- Evaluated model performance using precision, recall, F1-score, and top-k retrieval accuracy.

Honors and Achievements

2025 – Present	**Campus President**, AICP, Artificial Intelligence Community of Pakistan.
2025	**Co-Lead**, MSLA, Microsoft Student Learning Ambassadors.
2024	**Campus Ambassador**, Notion.
2023	**Campus Ambassador**, GDSC, Google Developer Student Clubs.
2021	**PEEF Scholarship**, HSSC, Punjab Educational Endowment Fund.
2021	**Dalda Scholarship**, HSSC, Merit-Based Award.
2019	**PEEF Scholarship**, SSC, Punjab Educational Endowment Fund.

Technical Skills

LLMs	Qwen3-1.7B, Llama 3.2-1B, BioBERT, Hugging Face Transformers, Ollama
NLP	Named Entity Recognition, TF-IDF/BM25, Semantic Search, Text Classification, spaCy
Speech & Audio	faster-whisper, SpeechBrain, Speaker Diarization, CTranslate2
Retrieval & RAG	FAISS, Sentence Transformers, Vector Search, PubMed API
Bioinformatics	Biopython, Sequence Alignment, Genomic Data Analysis, NCBI/PubMed Databases
Backend & APIs	FastAPI, WebSockets, SQLAlchemy, PostgreSQL, PyJWT
Frontend	React, TypeScript, Vite, axios
ML/Data Frameworks	PyTorch, scikit-learn, NumPy, pandas
DevOps	Docker, Nginx, Git, GitHub
Lang & Editors	Python, R, JavaScript, TypeScript, VS Code, Jupyter Notebook

Language Proficiency and Relevant Courseworks

English	Fluent (Professional Working Proficiency)
Urdu	Native
Coursework	Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems
Coursework	Machine Learning, Bioinformatics Computing, Systems Biology, Computational Genomics

Recent Projects

2023 **Protein 3D Structure Visualization**.

Tools: Biopython, RCSB PDB API, py3Dmol, Django

- Built an interactive tool to fetch and parse PDB structures using Biopython and the RCSB PDB REST API.
- Calculated secondary structure via DSSP and engineered geometric features like radial distance and angles.
- Rendered interactive 3D molecular visualizations using py3Dmol with cartoon, stick, sphere, and surface styles.

2024 **Differential Gene Expression Analysis of SCFA Treatment**.

Tools: GEOparse, scikit-learn, SciPy, Seaborn

- Built a transcriptomic pipeline using GEOparse to retrieve and process public GEO expression datasets.
- Applied parametric and non-parametric statistical testing with multiple-testing correction and Cohen's D.
- Clustered expression patterns using PCA and K-Means, and visualized results via volcano plots.

2025 **Scene2026: Gamified Year-Long Habit Tracker**.

Tools: React, Firebase, Tailwind CSS, Vite

- Designed a 4-level nested data architecture using a custom weekday-based calendar-generation algorithm.
- Implemented real-time data sync using Firebase Firestore listeners with anonymous and token-based auth.
- Built a custom rich-text editor using contentEditable with regex-based task completion parsing.

Present **Vehicle Detection using YOLOv8**.

Tools: YOLOv8, Ultralytics, OpenCV, Google Colab

- Trained a YOLOv8n model using mixed-precision (FP16) training for GPU memory and speed optimization.
- Built a synthetic dataset generation pipeline with automated YOLO-format bounding box labeling.
- Benchmarked detection performance using mAP50, precision, and recall on a held-out validation split.

Present **Malaria Parasite Detection System**.

Tools: YOLOv11, SAHI, Docker, FastAPI, Streamlit

- Engineered a synthetic data pipeline using Poisson blending to generate labeled blood smear training images.
- Fine-tuned YOLOv11 for real-time parasite localization using automated bounding box annotation.
- Integrated SAHI for tiled high-resolution inference and deployed via Dockerized FastAPI and Streamlit.